27 A green laser pointer and a red laser pointer of the same power cast circular images on a white screen of the same diameter. The wavelengths of the green and red lasers are λ_G and λ_R respectively.



If the pressure exerted by the green laser on the screen is p, what is the pressure exerted by the red laser?

A p **B** $\frac{\lambda_G}{\lambda_R}p$ **C** $\frac{\lambda_R}{\lambda_G}p$ **D** $\left(\frac{\lambda_G}{\lambda_R}\right)^2 p$

Ans: **A**

Momentum of photon = $\frac{h}{\lambda}$, Energy of photon = $\frac{hc}{\lambda}$

Rate of emission of photons = $\frac{P}{hc/\lambda}$ where P is power

Assuming all photons are reflected,

Force exerted on photons = $\frac{P}{hc/\lambda}\left(\frac{h}{\lambda}-\left(-\frac{h}{\lambda}\right)\right)=\frac{2P}{c}$ (N2L)

= Force on screen (N3L)

Pressure = $\frac{\frac{2P}{c}}{\pi\frac{D^2}{4}}=\frac{8P}{c\pi D^2}$ where D is diameter of image

Since pressure is independent of λ, and both lasers have the same power and produce images of the same diameter, pressure by red laser is also p.